

ORIGINAL RESEARCH

Optimal scheduling and management of grid-connected distributed resources using improved decomposition-based many-objective evolutionary algorithm

Ghulam Abbas¹  | Zhi Wu¹  | Aamir Ali² 

¹School of Electrical Engineering, Southeast University, Nanjing, China

²Department of Electrical Engineering, Quaid-e-Awam University of Engineering Science and Technology, Nawabshah, Sindh, Pakistan

Correspondence

Zhi Wu, School of Electrical Engineering, Southeast University, Nanjing 210096, China.
Email: zwu@seu.edu.cn

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Abstract

This paper emphasizes the integration of wind and photovoltaic (PV) generation with battery energy storage systems (BESS) in distribution networks (DNs) to enhance grid sustainability, reliability, and flexibility. A novel multi-objective optimization framework is introduced in this study to minimize energy supply costs, emissions, and energy losses while improving voltage deviation (VD) and voltage stability index (VSI). The proposed framework comprising normal boundary intersection (NBI) and decomposition-based evolutionary algorithms (DBEA) determines the optimal siting and sizing of renewable-based distributed resources, considering load demand variations and the intermittency of wind and solar outputs. The comparative analysis establishes that the proposed strategy performs better than many contemporary algorithms, specifically when all the objective functions are optimized simultaneously. The validation of the proposed framework was carried out on the standard IEEE-33 bus test network, which demonstrates significant percentage savings in energy supply costs (49.6%), emission rate (62.2%), and energy loss (92.3%), along with enormous improvements in VSI (91.9%) and VD (99.8953%). The obtained results categorically underline the efficiency, reliability, and robustness of the proposed approach when employed on any complex distribution network comprising multiple renewable energy sources and battery storage systems.

1 | INTRODUCTION

1.1 | Background, scope, and challenges

Massive carbon emissions are wreaking havoc around the world. Global warming and climate change have been making the lives of humans very difficult, and their impacts are seen in the shape of rapid glaciers melting, rising temperatures, and increasing turbulences in aircraft. The major cause of these hazards is burning fossil fuels; hence, it is imperative to shift traditional power systems to renewable energy sources completely. The abundant energy can be exploited through solar, wind, and water-based power generators. Distributed resources (DR) power systems are not directly connected to a bulk power transmission system; nonetheless, they include both distributed generations (DGs) and battery energy storage systems (BESS) which are vital components in the current development of

modern power systems [1]. Their collective impact on sustainability, reliability, and flexibility aligns seamlessly with the broader objectives of transitioning towards cleaner and more resilient energy infrastructures. As technological evolutions enhance, the critical roles played by DG and BESS are expected to expand, contributing significantly to a more sustainable and adaptive energy landscape [2, 3]. Due to the increasing demand for electricity and alarmingly rapid fossil fuel depletion, alternative energy sources have gained enormous attention in recent decades. Consequently, research into integrating distributed generations with distribution networks (DNs) has received extensive attention [4]. Furthermore, placing a DG in the best location with a suitable size can mitigate the necessity of reactive power to improve the voltage profile and reduce active power losses. Numerous researchers have proposed multiple optimization strategies to solve the formulated problems, which include conventional, artificial intelligence, and hybrid

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intelligent systems to solve the optimization problems [5]. Previously carried out studies on the optimal DGs' integration in the distribution networks have only focused on the minimization of power losses and included a single objective function employing analytical approaches in [6], heuristic and meta-heuristic methods that comprise genetic algorithm [7], ant lion optimization algorithm [8], and mixed integer non-linear programming [9]. The recent research involves several objective functions to optimize DG siting and sizing, including higher voltage stability and reduced operational and carbon dioxide emissions costs. Genetic algorithm-based methods BSOA [10] and GA [11], computational methods ICA [12], MNLP [13], and ALOA [9], and hybrid optimization methods Fuzzy [14], and HPSO [15] are employed to achieve the projected outcomes. Nevertheless, these approaches suffer heavily while optimizing multiple objective functions simultaneously, and the usage of weighted aggregation also makes the objective functions consume a long time to solve the problem and obtain desired solutions. In the next attempt, Pareto optimality is employed to solve the optimization problems with various objective functions. Given that the multiple objective functions are optimized uniformly, the Pareto optimality techniques yield Pareto sets of the best solutions to achieve optimal results. However, the primary method only generates a single outcome whenever the value of each objective is selected. Recent studies have exploited multi-objective evolutionary algorithms based on the Pareto optimality concept to find the optimal locations and sizing of DGs comprising NSGAI [16], INSGAI [17], IMOHS [18], MOShBAT [7], and non-dominated sorting stochastic fractal search (NSSFS) [19], in an attempt to avoid limitation. These employed evolutionary algorithms (EAs) render a choice of Pareto-optimal solutions from which the network can make a fair decision. The integration of multiple distributed generations simultaneously is typically analysed individually. However, combining these two sub-problems will be a better choice for the whole power system. Most authors optimize a single objective function to minimize power loss in distributed networks. Researchers in [20–22] pointed out that a minimal amount of power loss was found when the multiple DGs were combined and optimized to find an optimal solution. However, a few researchers have attempted to optimize such complex problems with additional objectives. In [23], the authors employed fuzzy-ACO (ant colony optimization) based on Pareto optimality on a single solar-based DG and DSTATCOM to find their optimal locations and sizes aimed to minimize active power loss, increase feeder loading balance balancing, and improve the voltage profile of the system. Since the above-discussed approaches evaluated the weighted sum of the objective functions, the Pareto optimality could not optimize the objective functions evenly. Hitherto, deliberate research studies treated the DG models as deterministic; however, practically, they are highly intermittent and volatile; specifically, distribution generations based on renewable energy. Most previously discussed literature focused on the optimal placement of distributed generations for a single period only. The allocation of renewable energy-based DGs is central in distribution networks because these DGs are environmentally friendly, reduce losses, enhance economic growth, and are free from generating greenhouse gases (GHG).

Hence, accepting their immense significance and benefits, societies and utility companies strive towards cleaner and more resilient energy systems. However, the described energy sources are highly intermittent and unpredictable and lead to several scheduling problems; the employment of battery energy storage systems (BESSs) assists in tackling the challenges caused by the intermittency of the renewable energy sources and demand and supply mismatches in the power network.

1.2 | Literature review, contributions, and structure of the paper

Battery energy storage systems are essential in managing and optimizing the utilization of renewable energy in distribution networks. They guarantee a steady and reliable power supply by storing surplus energy throughout a high-generation period and discharging it during a demand span. It reduces power variations, keeps grid stability, and plummets the necessity for costly power sources. Battery energy storage systems support power systems in emergencies and variations by enhancing power quality and supplying auxiliary power. Through this action, the effectiveness and reliability of renewable energy are improved, which leads to tangible economic benefits. An all-inclusive comparison of the proposed algorithm with the latest research studies is shown in Table 1.

Table 1 expounds a wide-ranging approach to the proposed algorithm, optimizing several objective functions together, which are solved for the first time, and such analysis does not exist in the literature review. The proposed algorithm successfully integrates distributed resources (DRs) comprising both distributed generation (DG) and battery energy storage systems (BESS). It solves all the objective functions simultaneously to achieve optimal solutions involving both deterministic and probabilistic methodologies in single-period and multi-period scenarios. The proposed framework effectively solves the formulated objective functions, that is, the cost of energy supplied (C), emission rate (E), energy loss reduction (CEL), voltage stability index (VSI), and voltage deviation (VD). In a comparative analysis, the results of the employed technique outperform many contemporary optimization algorithms by solving complex challenges related to distribution networks and validating their efficiency and superiority. Henceforth, this study makes massive contributions by significantly filling the research gap by solving five critical conflicting objective functions to optimize distributed resource allocation in a complex distribution network. According to the authors' best knowledge, a novel framework is introduced in this paper, which proposes a unified strategy to solve the multiple conflicting objective functions together in a multi-period time frame, which does not exist in the previous research works.

The presented study focuses on the critical role of renewable energy-based DGs and BESS integration and optimization in enhancing the economic and environmental performance of distribution networks. It introduces a dynamic, multi-objective framework to address the problems renewable energy sources posed by their intermittent and volatile nature. The proposed Improved Decomposition Based Evolutionary Algorithm

TABLE 1 An in-depth comparison of the proposed algorithm with the relevant latest research studies.

Method/Reference	S/M	DR type							B_{ESS}	t_1	t_n	d/p	OF
		Non-dispatchable			Dispatchable								
		P	Q	PQ	P	Q	PQ						
SPEA2/[23]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	PL, C, E	
BONMIN/[19]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	PL	
IBPSO/[24]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	C, CEL	
GWO/[25]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	PL	
EAE0/[26]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	PL, VD, VSI	
EGA/[27]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	PL,QL,VD	
JSA/[28]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	C, PL, QL, VD, VSI	
Cf-PSO/[29]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	C, PL, QL, VD, VSI	
GWO/[30]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	PL, QL, VD, VSI	
DE/[31]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	VD, PL, VSI	
EGWO-PSO/[14]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	C, E, PL, VD	
ALO/[32]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	C, PL, VD	
PSO/[33]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	PL, VD	
WCA/[34]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	C, PL, VD, VSI	
MRFO/[35]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	PL, VD, VSI	
WOA/[36]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	PL,VD	
MINLI-ECM/[37]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	C, E	
GA-DE-SPEA2/[38]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	C, E, VSI	
MSLF/[18]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	C, CEL, PL	
MALO/[8]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	PL, QL,VD, VSI	
MOCMIEP/[39]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	PL, E, VSI	
IGWOPSO/[40]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	PL,VD, VSI	
MOMVO/[41]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	C,VD	
MOEAD/[42]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	PL, QL	
I-DBEA/[43]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	PL,VD, VSI	
CSSA/[44]	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	PL, VD, VSI	
Proposed	☑/☑	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑/☑	C, E, CEL, VSI, VD	

Abbreviations: S, single objective; M, multi-objective; DR, distributed resources; B_{ESS} , battery energy storage system; t_1 , single period; t_n , multi-period; d, deterministic; p, probabilistic; OF, objective function; VSI, voltage stability Index; P, active power; Q, reactive power; PQ, active and reactive power; PL, active power losses; QL, reactive power losses; C, cost of energy supplied; E, Emission rate; CEL, cost of energy loss; VD, voltage deviation; VSI, voltage stability index.

(I-DBEA) framework demonstrates its substantial improvements by successfully solving the formulated problems and maintaining their validation, efficiency, and robustness. This paper substantially contributes to cost reduction, mitigation of energy losses, and voltage stability by optimizing the siting and sizing of renewable energy-based DGs and BESS devices while enhancing the reliability of the distributed network. Hence, it provides promising and comprehensive solutions for over 72 hours while incorporating real-time load curves and local climate data. The comparison of the proposed algorithm with the other existing multi-objective evolutionary algorithms (MOEAs) validates its superior convergence, diversity, and quality performance in handling complex power system planning problems. The main contributions of the paper are:

- The proposed framework integrates wind, solar PV, and BESS to reinforce grid sustainability, reliability, and flexibility.
- It develops a robust multi-objective optimization framework that addresses conflicting objectives, including cost minimization, emissions reduction, and voltage stability.
- Normal boundary intersection (NBI) and decomposition-based evolutionary algorithms are employed together to determine the optimal siting and sizing of distributed resources (DRs).
- The proposed technique is validated on a standard IEEE 33-bus network, showcasing its robustness in saving energy costs and significant improvements in voltage stability.

by using KCL at each bus as:

$$\underbrace{\begin{bmatrix} B_1 \\ B_2 \\ B_3 \\ B_4 \\ B_5 \end{bmatrix}}_{[B]} = \underbrace{\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}}_{[BIBC]} \times \underbrace{\begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \end{bmatrix}}_{[I]} \quad (6)$$

However, the voltage drop at each bus (forward sweep) for the reference bus is calculated using Kirchhoff's voltage law (KVL) as:

$$\underbrace{\begin{bmatrix} V_1 \\ V_1 \\ V_1 \\ V_1 \\ V_1 \end{bmatrix}}_{[\Delta V]} - \underbrace{\begin{bmatrix} V_2 \\ V_3 \\ V_4 \\ V_5 \\ V_6 \end{bmatrix}}_{[BCBV]} = \underbrace{\begin{bmatrix} Z_{12} & 0 & 0 & 0 & 0 \\ Z_{12} & Z_{23} & 0 & 0 & 0 \\ Z_{12} & Z_{23} & Z_{34} & 0 & 0 \\ Z_{12} & Z_{23} & Z_{34} & Z_{45} & 0 \\ Z_{12} & Z_{23} & 0 & 0 & Z_{36} \end{bmatrix}}_{[BIBC]} \times \underbrace{\begin{bmatrix} B_1 \\ B_2 \\ B_3 \\ B_4 \\ B_5 \end{bmatrix}}_{[B]} \quad (7)$$

Substitute the value of matrix [B] from Equation (6) in Equation (7) we get:

$$[\Delta V] = \underbrace{[BCBV] \times [BIBC]}_{DLF} \times [I] = [DLF] \times [I] \quad (8)$$

DLF is the distribution load flow matrix that helps calculate voltage drop at each bus w.r.t reference bus. The DLF matrix is calculated simply by using the following two steps:

- Step 1: Set branch impedance vector Z_b from the branch data.
- Step 2: Convert the branch impedance vector into the diagonal matrix, in which all the upper and lower elements are set to zero except the main diagonal as $Z = \text{diag}(Z_b)$ and multiply with [I] for ΔV .

2.2 | The proposed objective functions

This paper uses various objective functions to trace the proposed decision variables. The selection of objective functions is based on economical, technical, and reliability points of view.

2.2.1 | Cost of energy supplied (C)

The cost of energy supplied given in Equation (9) is considered one of the major objective functions in most research studies, as shown in Table 1.

$$C = \sum_{t \in T_n} \left(C_{SS}^t (P_{SS}^t) + \sum_{i \in N_{DG}} C_{DG_i}^t (P_{DG_i}^t) - \sum_{j \in N_{BESS}} C_{BESS_j}^t (P_{dch_j}^t) \right) \quad (9)$$

where indices t , i , and j show the operating indices of time interval, DGs, and BESS, respectively. T_n , N_{DG} and N_{BESS} are the total number of time slots, DGs, and BESS, respectively. C_{SS} , C_{DG} and C_{BESS} are the cost of power supplied by the upstream substation, DG, and BESS, respectively. The cost parameters of active power supplied by substation, DGs, and BESS are measured in \$/MWh.

$$C_{SS}^t (P_{SS}^t) = k_1 \times P_{SS}^t \quad (10)$$

$$C_{DG}^t (P_{DG}^t) = k_1 \times (P_{DG}^t)^2 + k_2 \times P_{DG}^t + k_3 \quad (11)$$

$$C_{BESS}^t (P_{dch}^t) = k_1 \times (P_{dch}^t)^2 + k_2 \times P_{dch}^t + k_3 \quad (12)$$

Here C_{DG} , C_{BESS} , and C_{loss} are the cost of DGs, the discharging cost of BESS, and the cost of energy losses, respectively while k_1 , k_2 and k_3 are the variable quadratic cost coefficients. Whereas, C_{SS}^t , $C_{DG_i}^t$ are the cost parameters of power supplied by the substation and DGs, respectively.

2.2.2 | Emission (E)

The emissions, including CO₂, SO₂, and NO_x from the generation units, are considered to be the most effective pollutants in power generation sources. The mathematical formulation of this objective function can be described as follows:

$$E = \sum_{t \in T_n} \left(E_{Grid}^t + \sum_{i=1}^{N_{DG}} E_{DG_j}^t \right) \quad (13)$$

whereas,

$$E_{DG_j} = (CO_2^{DG} + NO_x^{DG} + SO_2^{DG}) \times P_{DG_j} \quad (14)$$

$$E_{Grid} = (CO_2^{Grid} + NO_x^{Grid} + SO_2^{Grid}) \times P_{gGrid} \quad (15)$$

Whereas P_{gGrid} is the active power supplied by the grid station. In our case, only non-dispatchable types of DGs are integrated; therefore, in this study, the emission rate of DGs is set to $E_{DG_i} = 0$.

2.2.3 | Cost of energy loss (CEL)

The cost of energy loss (CEL) in the entire time horizon is given as:

$$CEL = \sum_{t \in T} (\alpha^t (P_{loss}^t)) \quad (16)$$

Whereas α^t is the co-efficient of power loss equal to 80.49\$/MWh [45] while P_{loss} is the total active power loss and can be calculated as:

$$P_{loss} = \sum_{i=1}^N |B_i|^2 \times R_i = [R]^T [BIBC] \times [I]^2 \quad (17)$$

where B_l and R_l are the l^{th} branch current and resistance, respectively. $[R]$ is the branch resistance matrix containing the resistance of all the branches. While the current is decomposed into its real and imaginary parts as:

$$P_{\text{loss}} = [R]^T \times \left(([BIBC] \times \text{re}(I))^2 + ([BIBC] \times \text{Im}(I))^2 \right) \quad (18)$$

By substituting the Equation (5) in Equation (18), the total active power losses (P_{loss}) can be calculated as:

$$P_{\text{loss}} = [R]^T \left([BIBC] \left[\frac{P \cos(\theta) + Q \sin(\theta)}{|V|} \right] \right)^2 + [R]^T \left([BIBC] \left[\frac{P \cos(\theta) + Q \sin(\theta)}{|V|} \right] \right)^2 \quad (19)$$

2.2.4 | Voltage deviation (VD)

Nodal voltage magnitude is important to evaluate the security and power quality of the distribution network. Minimizing voltage deviation (VD) can help to guarantee better voltage levels in the power distribution systems. Therefore, the third objective function is related to the security of the distribution network, which is achieved by minimizing voltage deviation between each load bus and reference bus and can be calculated as:

$$\text{VD} = \sum_{t \in T} \left(\sum_{m \in N_b} |V_{\text{SS}} - V_m|^2 \right) \quad (20)$$

where m is the particular bus number, and N_b are the total number of buses in the network. V_{ref} is the voltage at slack bus, and it is set to 1 p.u in this paper while V_m is the voltage at a specific bus.

2.2.5 | Voltage stability index (VSI)

The security level of a distribution system is insufficient if voltage deviation is considered alone. Therefore, to have a better voltage profile, the VSI [46] is proposed as one of the objective functions of this study. The maximum VSI of the distribution system means the bus can preserve its voltage profile within satisfactory limits over variable loading conditions. For the secure distribution system operation, the utility entities try to maintain the VSI of the distribution system near to the unity for all buses. It can be calculated as:

$$\text{VSI}_k^t = \left[V_f^t \right]^4 - 4 \left(P_{Lk}^t r_l + Q_{Lk}^t x_l \right) \left[V_f^t \right]^2 - 4 \left(P_{Lk}^t x_l - Q_{Lk}^t r_l \right)^2 \quad (21)$$

where, V_f^t is the sending end voltage at any time t . P_{Lk} , and Q_{Lk} , are the active and reactive power injections at receiving end bus k . r_l and x_l are the line resistance and inductive reactance, respectively, a link between sending end bus f and receiving

end bus k . During operation, a bus with the lowest value of VSI among all others needs to be maximized to enhance voltage level of the entire network. Consequently, to achieve the maximum VSI, the objective function VSI can be formulated as [43].

$$\text{VSI} = - \left(\min \left(\text{VSI}_k^t \right) \right) \quad (22)$$

2.3 | Constraints

2.3.1 | Constraints of equality and inequality

The equality constraints are the equations that ensure power balance in the network, where both generated active and reactive power must be equal to the load demand and losses in the DN.

$$\sum_{t \in T} \left(P_{\text{SS}}^t + \sum_{i \in N_{\text{ESS}}} P_{\text{dch}_i}^t + \sum_{i \in N_{\text{DG}}} P_{\text{DG}_i}^t - \sum_{b \in N_b} \left(P_{D_b}^t + R_{\text{loss}} \right) - \sum_{i \in N_{\text{ESS}}} P_{\text{ch}_i}^t \right) = 0 \quad (23)$$

$$\sum_{t \in T} \left(Q_{\text{SS}}^t + \sum_{i \in N_{\text{DG}}} Q_{\text{DG}_i}^t - \sum_{b \in N_b} \left(Q_{D_b}^t + Q_{\text{loss}} \right) \right) = 0 \quad (24)$$

where, P_{sub} and Q_{sub} represent the active and reactive power of the main sub-station. P_D and Q_D are the active and reactive demand, R_{loss} and Q_{loss} are the active and reactive power losses in the network. While $P_{\text{ch}_i}^t$ and $P_{\text{Dch}_i}^t$ represent the charging and discharging power of battery energy storage systems.

2.3.2 | Inequality constraints

The inequality constraints set limits on the different operating parameters of power systems and their components, as well as the security constraints of the line and load buses.

$$P_{\text{SS}}^{\min} \leq P_{\text{SS}} \leq P_{\text{SS}}^{\max}; Q_{\text{SS}}^{\min} \leq Q_{\text{SS}} \leq Q_{\text{SS}}^{\max} \quad (25)$$

$$P_{\text{DG}_i}^{\min} \leq P_{\text{DG}_i} \leq P_{\text{DG}_i}^{\max}; Q_{\text{DG}_i}^{\min} \leq Q_{\text{DG}_i} \leq Q_{\text{DG}_i}^{\max} \quad (26)$$

$$\sum_{i \in N_{\text{DG}}} P_{\text{DG}_i} \leq \sum_{j \in N_b} P_{D,j} \quad (27)$$

$$\sum_{i \in N_{\text{DG}}} Q_{\text{DG}_i} \leq \sum_{j \in N_b} Q_{D,j} \quad (28)$$

$$V_{\text{DG}_i}^{\min} \leq V_{\text{DG}_i} \leq V_{\text{DG}_i}^{\max} \quad (29)$$

$$V_m^{\min} \leq V_m \leq V_m^{\max} \quad (30)$$

$$S_1^t \leq S_1^{\max} \quad (31)$$

$$n_1 = N_b - 1 \quad (32)$$

$$\det(\mathbf{A}) = \begin{cases} 1 \text{ or } -1, & (\text{radial network}) \\ 0 & (\text{not radial}) \end{cases} \quad (33)$$

$$\text{SoC}_j^t = \text{SoC}_j^{t-1} + \frac{\eta_j P_{\text{dch}_j}^t}{E_j^{\text{max}}} - \frac{P_{\text{ch}_j}^t}{\eta_j E_j^{\text{max}}} \quad (34)$$

$$\text{SoC}_{j(\text{ini})} = \text{SoC}_j^{t-1} = 1.8 \text{ MWh} \quad (35)$$

$$0.1 * E_j^{\text{max}} \leq \text{SoC}_j \leq 0.9 * E_j^{\text{max}} \quad (36)$$

$$0 \leq P_{\text{ch}_j}^t \leq 1 \text{ MW and } 0 \leq P_{\text{dch}_j}^t \leq 1 \text{ MW} \quad (37)$$

Whereas SoC_i^t represents the state of charge (SoC) of the BESS for each time step. It should be noted that the charging and discharging power of the BESS units are kept identical to avoid any internal energy exchange among different storage units at each time interval [47, 48], while E_i^{max} represents the maximum capacity of the BESS set to be 1.8 MWh.

Equations (25) and (26) are the active and reactive power limits of the main substation identified in the reference bus of this study. Equations (27) to (28) describe the cumulative power constraints of DGs, ensuring that the total generated active and reactive power remains within or equal to the total load demand. Voltage constraints for the PV and PQ buses are addressed in Equations (29) and (30), respectively, while Equation (31) pertains to the MVA branch flow limit. Equations (32)–(33) establish the radiality constraints of the distribution system, ensuring its tree-like structure. The connection matrix and bus incident matrix play a pivotal role in network configuration, where $A_{ij} = 1$ indicates that the i^{th} branch is linked from bus j , $A_{ij} = -1$ signifies that the i^{th} branch is connected to bus i and $A_{ij} = 0$ shows that there is no connection between the i^{th} branch and j^{th} bus. Equations (34)–(37) outline the inequality constraints of the BESS, representing their intertemporal constraints linked to the various periods. The forward-backward sweep method calculates load flow Equations (23) and (24), ensuring all the equality constraints. On the other hand, the inequality constraints given in Equations (25)–(37) are solved using the representative epsilon constraint method. The proposed framework successfully formulates the problem posed by the distribution network while ensuring robust and efficient solutions.

2.4 | Decision vector

In this paper, the decision vector $\mathbf{x}$ is given in Equation (38), which comprises both continuous and integer variables:

$$\mathbf{x} = [\text{site}_{\text{DG,S}}, \text{size}_{\text{DG,S}}, \text{site}_{\text{DG,W}}, \text{size}_{\text{DG,W}}, \text{site}_{\text{BESS}}, \text{size}_{\text{BESS}}, V_{\text{DG}}] \quad (38)$$

The decision vector $\mathbf{x}$ given in Equation (38) comprises the site of wind type DG as $\text{site}_{\text{DG,W}}$, size of wind-type

DG as $\text{size}_{\text{DG,W}}$, site and size of solar PV type DG as $\text{site}_{\text{DG,S}}$, $\text{size}_{\text{DG,S}}$ respectively and V_{DG} represents the voltage set point of all generator buses.

3 | IMPLEMENTATION OF THE PROPOSED FRAMEWORK

The allocation of distributed resources is a multi-objective mixed integer non-linear problem. It poses a challenge to the existing multi-objective evolutionary algorithms due to their small feasible search space and constraints in both objective and control variables space. To address this lacuna, this paper proposes an improved decomposition-based evolutionary algorithm (I-DBEA) employed with the epsilon constraint method to handle constraints and solve highly complex multi-objective problems.

3.1 | Implementation of improved decomposition based evolutionary (I-DBEA) algorithm

Since the multi-period DR allocation is a mixed integer non-linear problem hence, it becomes extremely challenging for the MOEAs to achieve well-converged and evenly distributed Pareto Front (PF) solutions. In the literature, most of the existing MOEAs prioritize feasible solutions, which leads to two issues: Firstly, the population may become stuck in local optimal feasible regions. Secondly, since the population only grows on the feasible side, the search space may not be entirely explored. The flowchart of the proposed optimization technique is shown in Figure 2. The principal steps of the I-DBEA technique are discussed in the following sections.

3.2 | Generate the reference points

First, a W set of reference points is generated using the normal boundary intersection method [49], and these reference points collectively control the trade-off between exploration and exploitation. In this study, 153 reference points are generated using Equation (12), and the directions of reference points are modelled by building a straight line from the origin.

$$W = M^{+s-1} C_s \quad (39)$$

Hyperplane for the objective functions using the NBI approach with an evenly spaced of $\delta = 0.0625$ is calculated by using $\delta = 1/s$ where $s = 16$. The distribution of reference points for the three objective functions using a 3D normalized hyperplane is shown in Figure 3.

The initial randomly generated population size of the proposed method is set equal to the reference points within the range of decision vectors.

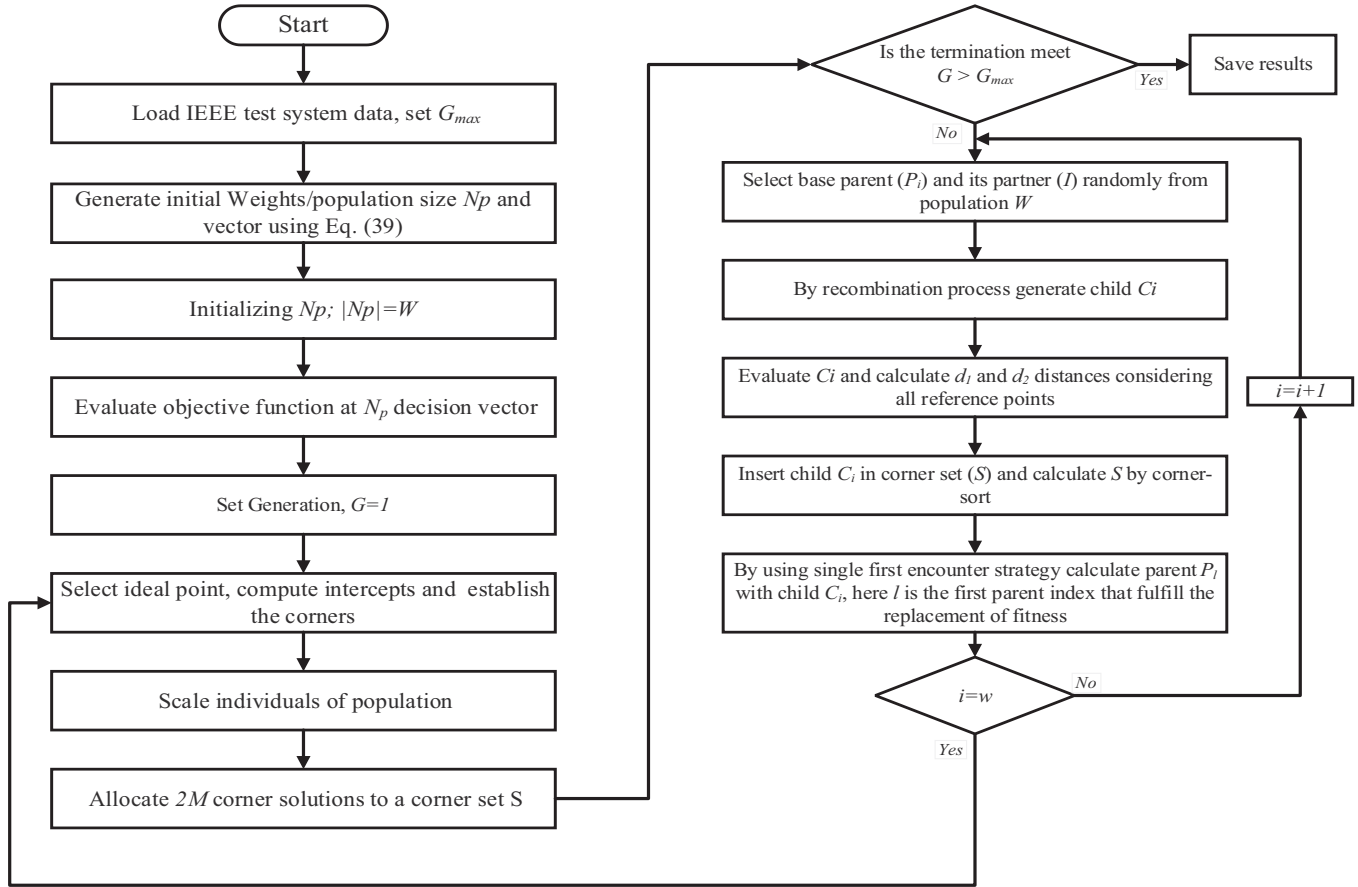


FIGURE 2 Flowchart of I-DBEA optimization method.

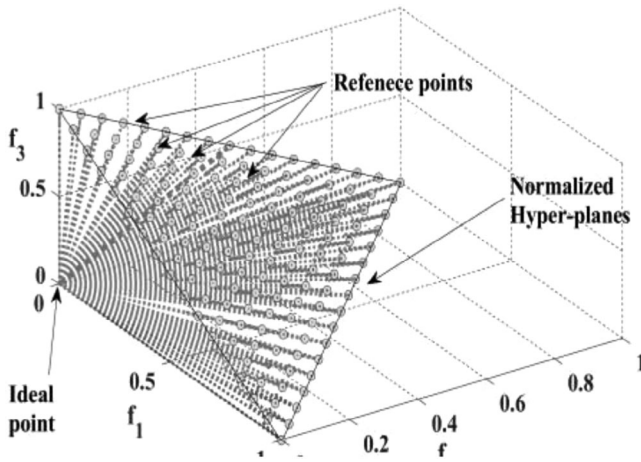


FIGURE 3 A set of reference vectors was generated using the NBI approach for three objective functions.

3.3 | Computation and normalization of distances

Usually, in a decomposition-based approach, the fitness of objective functions with respect to given directions is determined using two distance measurements, d_1 and d_2 , as expressed

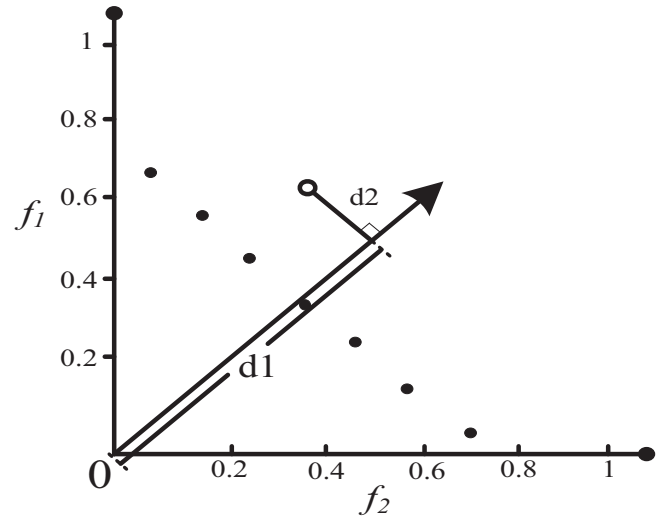


FIGURE 4 Measurement of distance w.r.t reference direction.

in Equations (40) and (41), respectively. These two measurements are shown in Figure 4 and calculated as:

$$d_1 = w^T f'(x) \quad (40)$$

$$d_2 = \|f'(x) - w^T f(x)w\| \quad (41)$$

where, w is the unit vector showing direction along a given reference point, d_1 is the Euclidian distance between the origin and footnote, and d_2 is the normal length of the reference path.

The convergence and diversity of the proposed technique can be controlled by adjusting the values of d_1 and d_2 . If $d_2 = 0$, then it shows perfect diversity, which means the solutions are ideally aligned throughout the length of the required reference route, whereas a lower value of d_1 gives superior convergence. In the proposed algorithm, $2M$ results are recognized through the corner sort rank method given in [50]. The topmost M solutions are the smallest in all objectives, whereas the following M solutions are founded using the L2-norm of all but one objective. From these solutions, the maximum value in each objective is recognized, and the corresponding solution is nominated and referred to as the extreme point $= z^e$. These points are employed to produce hypervolume and calculate the extreme value of the intercepted objective function. On the other hand, the population at which objective functions are minimum called ideal points, represented by $z = (f_1^{\min}, f_2^{\min}, \dots, f_M^{\min})$. The general equation of the plane through ideal points can be given as:

$$C_1 f_1 + C_2 f_2 + \dots + C_M f_M = 1 \quad (42)$$

Whereas $C_1, C_2, \dots, C_M$ are unit normal planes and can be defined as $C_1 = 1/a_1, C_2 = 1/a_2$, and $C_M = 1/a_M$. Where $a_1, a_2, \dots, a_M$ are the intercepts of hypervolumes along the objective function axes. Every optimized value of the objective functions can be scaled using the Equation (43).

$$f'_j(x) = \frac{f_j(x) - z_j}{a_j - z_j}, \forall j = 1, 2, \dots, M \quad (43)$$

3.4 | Method of recombination

In this section, a simulated binary crossover (SBX) technique proposed in [45] and polynomial mutation is employed to create two-child solutions. The first solution is achieved from these solutions as an individual trying to substitute any parent in the population.

3.5 | Replacement and selection

In steady state form, the child solution will enter the population through replacement if it is a non-dominated individual. The solution for child contests with all the solutions of the population in arbitrary order until it successfully replaces all the individuals. The value of distances $\{d_1, d_2\}$ represents the r^{th} solution in a randomly selected population and $\{d_1, d_2\}$ represents the child solutions in the r^{th} reference direction. If $d_{2c} < d_{2r}$ a child is considered as winner or $d_{2c} = d_{2r}$ and $d_{1c} < d_{1r}$ elsewhere d_{1r} and d_{2r} remain selected. The preference of d_2 over d_1 removes the necessity of complex ε -based arrangement.

3.6 | Handling of constraints

The handling of the constraint technique is based on ε level comparison described in [51]. The acceptable constraint violation (CV) can be calculated as:

$$CV = \sum_{i=1}^p \max(|g_i - \varepsilon|, 0) + \sum_{i=1}^q \max(b_i, 0) \quad (44)$$

$$CV_{mean} = \frac{1}{W} \sum_{j=1}^W (CV_j) \quad (45)$$

$$\text{Allowable violation, } (\varepsilon_{CV}) = CV_{mean} * FR \quad (46)$$

where, FR is the feasibility ratio between the feasible solutions to the size of the population (W).

4 | SIMULATION RESULTS OF DIFFERENT CASE SCENARIOS

This study optimizes a complex multi-objective, multi-period DR allocation problem employing an I-DBEA framework. The optimal site and size of renewable energy-based DGs and BESS are mixed integer decision variable problems. The non-dispatchable solar-based DGs are implemented to integrate active power only (operating at unity pf). However, wind-type DGs are integrated to locally inject variable reactive power that highly supports the voltage profile of the distribution system and decreases active power loss to some extent. The simulations of the formulated problem were conducted on the standard IEEE-33 bus network that was used worldwide, as demonstrated in Figure 5.

Three solar-based DGs and three wind-type DGs with each rated capacity of 1 MW were selected to be installed at the optimal locations of the proposed test network. The six-battery energy storage systems are also installed in the proposed test system with a maximum capacity of 2.7 MWh each, possessing a minimum and maximum charging and discharging capacity of 1 MW. The data involving the installation of DGs and BESS are extracted from [52, 53]. The proposed framework successfully schedules grid energy supply according to the rated power of DGs and load variations. The proposed problem was solved via the I-DBEA algorithm aimed to simultaneously optimize two, three, and five conflicting objective functions as described in the problem formulation section. This study considers three scenarios: steady-state single-period load, single-period comprised of various load models, and a real multi-period optimal DG and BESS allocation problem. To show the efficacy of the proposed strategy, it is duly compared with other contemporary algorithms deliberated in the literature review. The proposed case studies with different scenarios are given in Table 2.

The simulation results of each case scenario are detailed in the following sections based on the proposed objectives and Pareto front characteristics. The Pareto optimal solutions of the proposed bi-and tri-objective function are obtained via I-DBEA employment. The proposed framework successfully optimized

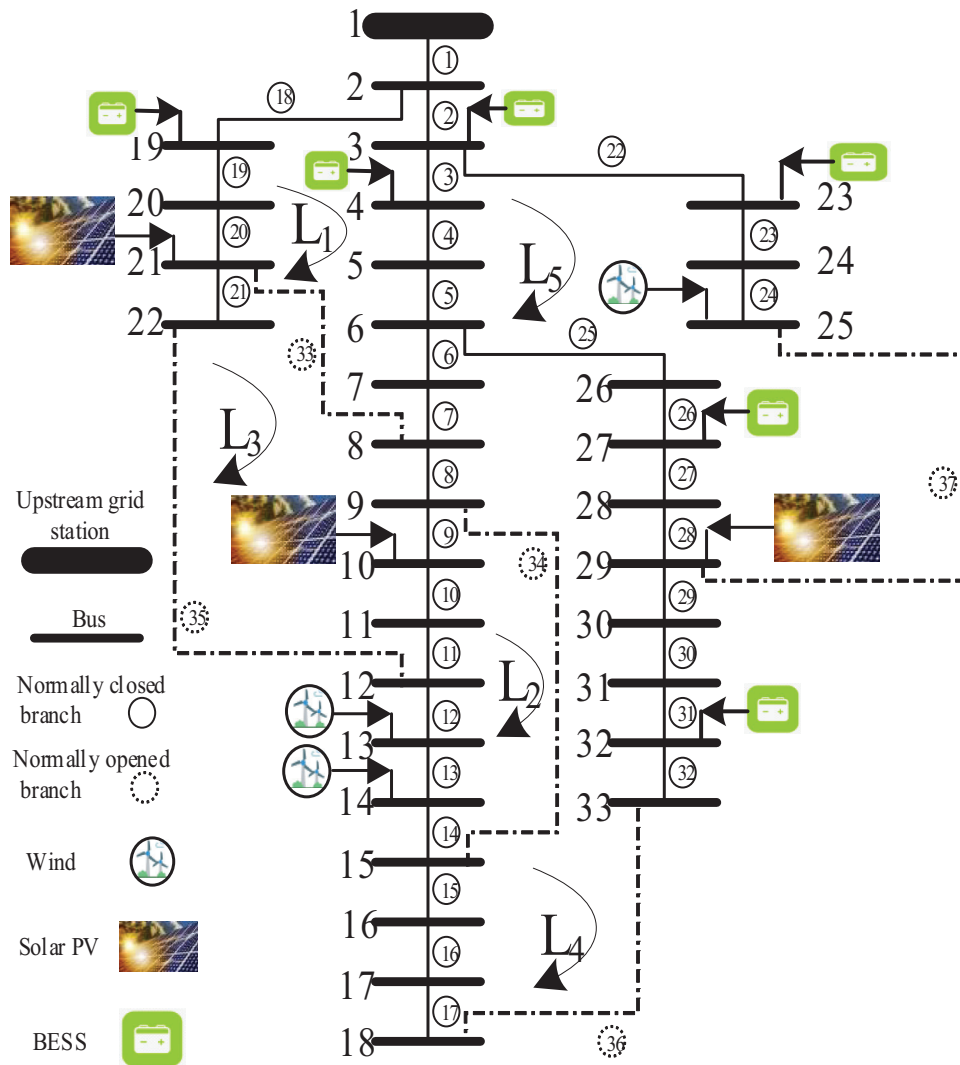


FIGURE 5 The base configuration of the proposed IEEE 33-bus network.

the proposed framework using MATLAB programming. All the simulations in this study were performed on the platform developed by Tian et al. (PlatEMO) [54]. All the codes were run on core i7 PC MATLAB 2021b version 9.10. The simulation results of three scenarios are analysed and discussed below to show the performance of the proposed algorithm against other recent MOEAs.

4.1 | Scenario 1. Single point LOA integrating DG only (unity and lagging PF)

Table 3 summarizes the simulation results of cases 1 and 2; the results of the proposed technique are compared with those of other optimization methods. Table 3 shows that the proposed method gives the most promising results for the multi-objective optimal site and size of the DG allocation problem with maximum DG integration compared to existing methods. The proposed framework also results in significant loss reduction

and voltage deviation, contrary to other contemporary algorithms [8, 49, 55–57]. Table 3 also shows that the teaching learning-based optimization (TLBO) [65], GA, PSO, and hybrid GA and PSO [66] are incapable of injecting the required amount of DGs to achieve the desired goal. Besides, the proposed framework enhances all the objective functions, that is, PL, VD, and VSI. The proposed I-DBEA technique reduces the active power losses from 202.667 to 94.8514 kW (53.21% reduction), voltage deviation from 0.1171 to 0.0007 p.u and enhances the VSI from 0.6988 to 0.965 p.u.

In case 1, the DGs cannot inject reactive power into the distribution network to optimize objective functions. In case 2, the proposed technique finds the optimal site and size of three DGs at 0.85 LPF. Table 3 shows that the proposed method outperforms other methods to achieve optimal site and size of DGs. The proposed algorithm minimizes power losses and voltage deviations and improves the voltage stability index than optimized in [57]. Table 3 also shows the improved analytical (IA) method. Ref. [26] and loss sensitivity factor simulated

TABLE 2 The proposed case studies with different scenarios.

# of Scenarios	Explanation	# of Objective functions	Case study	Explanation
1	Single point loading considering DGs only at unity and lagging pf	3	Case 1 (at unity pf)	PL vs. VD vs. VSI
		3	Case 2 (at 0.85 lag pf)	PL vs. VD vs. VSI
2	Single point loading considering simultaneous DGs and SCs	2	Case 1	PL vs. VD
		2	Case 2	PL vs. CEL
		3	Case 3	PL vs. CEL vs. VD
3	Impact of conflicting objective functions in multiperiod considering DG and BESS	2	Case 1	C_E
		2	Case 2	C_VSI
		2	Case 3	C_CEL
		2	Case 4	C_VD
		2	Case 5	E_VSI'
		2	Case 6	E_CEL
		2	Case 7	E_VD
		2	Case 8	VSI_CEL
		2	Case 9	VSI_VD
		2	Case 10	CEL_VD
		2	Case 11	C_E_VSI
		3	Case 12	C_E_CEL
		3	Case 13	C_E_VD
		3	Case 14	E_VSI_CEL
		3	Case 15	E_VSI_VD
		3	Case 16	VSI_CEL_VD
		5	Case 17	C_E_CEL_VSI_VD

TABLE 3 Simulation results of Case 1 (3 DGs operating at UPF) and Case 2 (3 DGs operating at 0.85 LPF).

Cases	Technique	Location	Cumulative DG size (MW)	RPL (KW) (% loss reduction)	VD (p.u)	VSI (p.u)
Case 1	Base case	—	—	202.667	0.1171	0.6988
	CTLBO [55]	13,24,30	3.7211	85.9595 (57.59%)	0.0026	0.9481
	TLBO [14]	12,28,30	3.5602	124.695 (38.47%)	0.0011	0.9503
	QOTLBO	13,26,30	3.4702	103.403 (48.97%)	0.0011	0.9530
	GA [58]	11,29,30	2.9940	106.3 (47.56%)	0.0407	0.9497
	PSO [58]	8,14,32	2.9890	105.3 (48.05%)	0.0335	0.9256
	GA/PSO [58]	11,16,32	2.9980	103.4 (48.99%)	0.0124	0.9508
	IMOEHO [57]	14,24,30	3.8520	95.0 (53.13%)	0.0008	0.9673
	Proposed	13,24,30	3.9131	94.8514 (53.21%)	0.0007	0.965
Case 2	IMOEHO [57]	13, 24, 30	3.5830	14.9 (92.6%)	0.0003	0.9814
	Analytical [59]	6, 25, 31	3.5800	26.1 (87.12%)	0.0086	0.8678
	LSFSA [60]	6, 18, 30	2.9980	26.7 (86.83%)	0.0013	0.9323
	IA [26]	6, 14, 30	2.8590	23.1 (88.60%)	—	—
	Proposed	13, 24, 30	3.0307	14.57 (92.8081%)	0.0002	0.9733

annealing (LSFSA) [29, 44] cannot inject enough DGs to achieve the desired goal, while the improved multi-objective elephant herding optimization (IMOEHO) [57] injects a lot of DG power. In the distribution network, the most significant reductions were observed in power losses and decreases by more than 92% in case 2. Besides, the voltage profile across all the buses approaches unity, and voltage deviations were reduced to a mere 0.0002 p.u. only.

4.2 | Scenario 2. Single point loading considering simultaneous DGs and SCs

The multi-objective evolutionary algorithms aim to find high-quality, non-dominated (ND) solutions in a simulation run. The quality measurement of the ND solutions has three goals, that is, convergence (minimum distance between the ideal point and PF), diversity (uniform distribution), and maximum spread, which should be considered while assessing the performance of MOEAs. The final obtained non-dominated MOEA solutions must have maximum convergence, diversity, and spread. Several criteria have been proposed to compare the performance of MOEAs precisely. This study measures the convergence and diversity of MOEAs using the hypervolume indicator (HVI) performance technique. In the HVI metric, a reference point (ideally around the nadir point) is obtained from the real PF, and no exact Pareto optimum solution exists in the real-world DG and SC allocation problem. Due to the absence of the real Pareto fronts, the exact nadir point remains to be determined. The optimal and nadir locations are obtained by executing each approach 20 times separately. Afterward, all the ND solutions are combined, and dominated solutions are found via ND sorting techniques [36]. The real Pareto optimum set combines the filtered set of ND solutions to carry out optimal and nadir values of MOEA. The best PF on a problem is the one with the biggest HVI. Four DG and DG-SC study cases of small to large-scale radial distribution systems with constant PQ load models are studied to evaluate the performance of the proposed algorithm with the latest MOEAs. The parameters were kept the same for the proposed and other contemporary algorithms. The population size N is kept at 50, and the number of objective functions (M) and maximum iterations (G) are given in Table 3.

Table 4 shows the statistical performance of the proposed framework based on HVI compared to the recently developed MOEAs (NSGAII [36], CAEAD [61], CCMO [62], ToP [63], CMOEA-MS [64], cDPEA [65]). No constraint violations were observed during the simulations of the case studies. Simulation findings in Table 4 show that the proposed approach outperforms or is approximately similar to the existing algorithms based on best and worst HVI values in most cases. Table 4 shows that the dark and highlighted results are the best, while half-dark are counted on the second-ranking. The proposed algorithms outperform and appear the best on most HVI tests. Table 4 shows that the I-DBEA algorithm has the best HVI values in all test cases except case 3 with 33-bus only DG. As shown in Table 4, CAEAD, CMOEA_MS, and cDPEA did not perform well in all cases on both test systems. However, 33-

bus DG-SC allocation yields the second-best values for cDPEA. The CCMO algorithm outperforms only in case 3 of 33-bus DG, while the proposed algorithm proves its superiority almost in cases. Table 4 indicates that the proposed approach excels when compared to other MOEAs. From Table 4, it can also be noticed that the ToP also fails to achieve the global PF.

Figure 6 shows the final non-dominated populations of the best PFs based on the HVI for each algorithm. Pareto fronts shown in Figure 6 prove that the non-dominated solutions are usually convergent and uniformly distributed. The PF of the ToP algorithm gets trapped into the local optima in case 1 of the DG-SC allocation problem, while the PF of the proposed algorithm is found to be well distributed. It can be seen in Figure 6 that the proposed algorithm can find widely spread, evenly spaced, non-dominated solutions; hence, it effortlessly tackles the optimal DG site and size and DG-SC allocation problems.

4.3 | Scenario 3. Impact of conflicting objectives on optimal distributed resource allocation considering multi-period study

This section solves a complex multi-period DR allocation problem using the proposed algorithm. The earlier section showcased the capabilities of the algorithm through two distinct scenarios. The first case analysed DG performance within a constant PQ load model. In contrast, the second case broadened the scope by examining the effects of integrating both DG and shunt capacitors (SCs), focusing on single-point loading problems in small-scale distribution systems to ascertain their optimal locations and sizes. This section also focuses on solving a complex multi-period large-scale DR allocation problem, validating the proposed technique's robustness and efficiency. The complexity of the problem involves 744 decision variables across various cases to test the ability of the proposed algorithm to tackle the dynamic challenges posed by the optimization problems.

Figure 7 vividly demonstrates the trade-off between two and three conflicting objective functions, showing the optimization problem's outcomes.

Figure 7 shows how the proposed algorithm constantly achieves convergence and generates evenly distributed non-dominated solutions. A significant aspect of the proposed optimization is its capability to span the entire temporal sequence in a single simulation run, underscoring the efficiency and robustness of the algorithm in tackling DR allocation problems. Figure 7 shows the final non-dominated solutions of the first four cases where the proposed algorithm succeeds in finding optimal solutions. The best compromise solution (BCS), highlighted in a large circle in Figure 7, signifies the most balanced solution among the conflicting objectives, encapsulating the efficiency of the proposed algorithm in all cases. Figure 7 illustrates the Pareto fronts of all four cases. It can be noticed from Figure 7 that the PFs for case 1, case 2, and case 3 are not smooth, which implies that a complicated relationship exists among objectives being optimized. In case 1, the trade-off

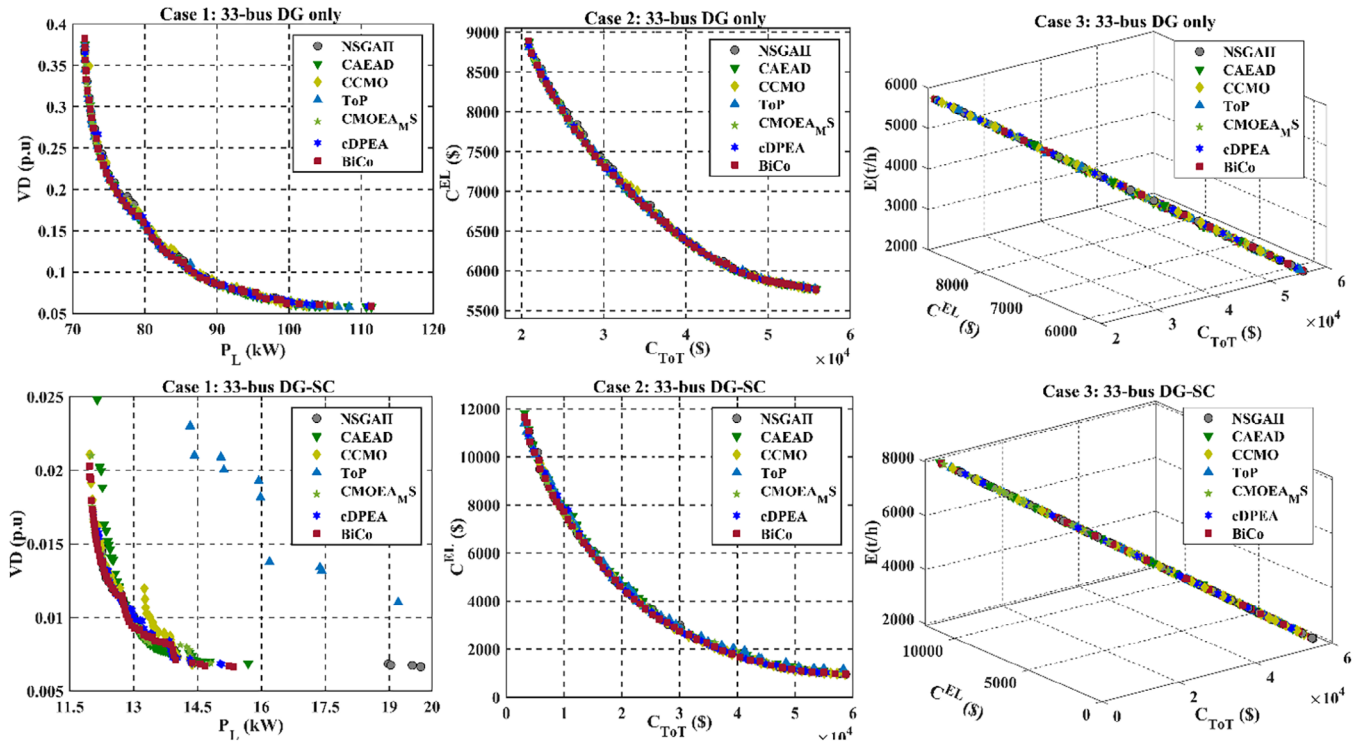


FIGURE 6 Final non-dominated solutions of state-of-art EAs vs proposed algorithm for the optimal site and size of DG and SC.

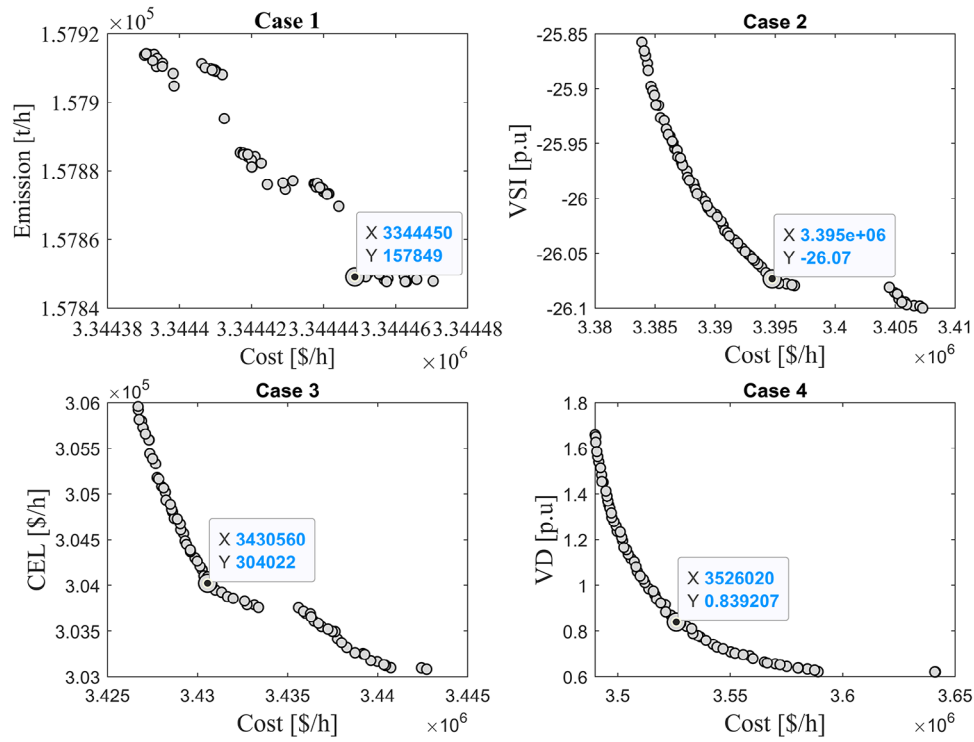
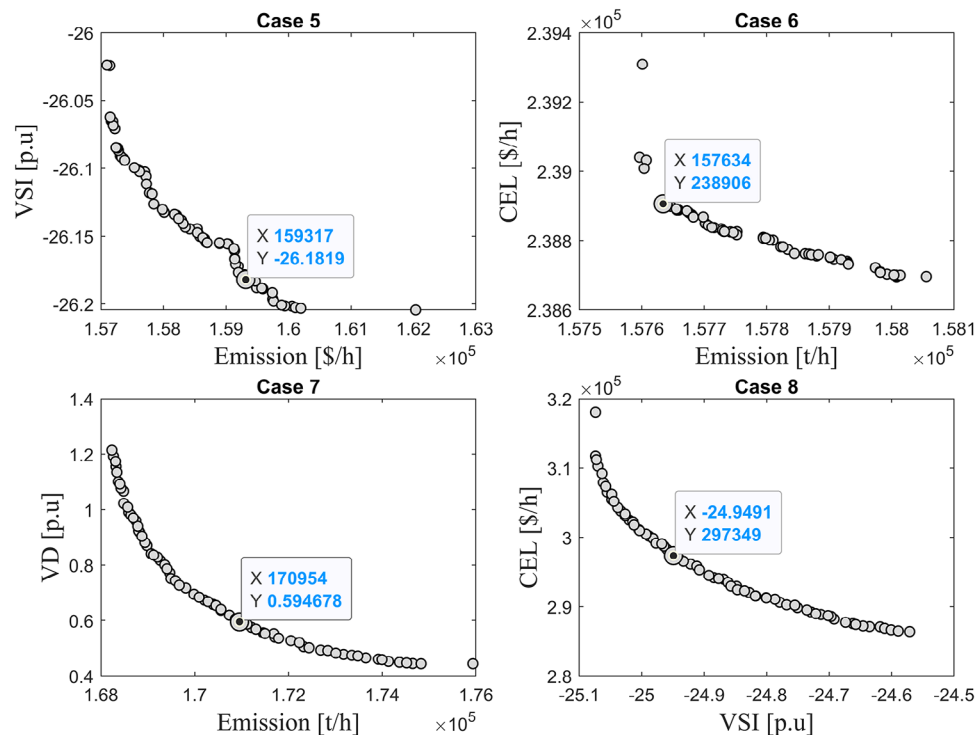


FIGURE 7 The final non-dominated solutions of cases 1–4 solve different bi-objective functions.

TABLE 4 Statistical results of best and worst values of HVI performance metric.

Test case	Problem	<i>M</i>	<i>G</i>	name	Proposed	NSGAI	CAEAD	CCMO	ToP	CMOEA_MS	cDPEA
33-bus only DG	Case 1	2	500	Best	0.31	0.31	0.31	0.31	0.31	0.31	0.31
				Worst	0.31	0.31	0.31	0.31	0.31	0.16	0.31
	Case 2	2	600	Best	0.21	0.21	0.21	0.21	0.21	0.21	0.21
				Worst	0.21	0.21	0.21	0.21	0.21	0.21	0.21
	Case 3	3	800	Best	0.10	0.10	0.09	0.10	0.09	0.10	0.10
				Worst	0.10	0.09	0.09	0.10	0.09	0.09	0.09
33-bus DG-SC	Case 1	2	500	Best	0.25	0.25	0.25	0.25	0.09	0.25	0.25
				Worst	0.25	0.22	0.22	0.22	0.00	0.24	0.24
	Case 2	2	600	Best	0.68	0.68	0.67	0.68	0.67	0.66	0.68
				Worst	0.68	0.66	0.66	0.67	0.63	0.65	0.67
	Case 3	3	800	Best	0.35	0.34	0.33	0.35	0.32	0.33	0.35
				Worst	0.34	0.33	0.32	0.34	0.30	0.31	0.33

**FIGURE 8** The final non-dominated solutions of cases 5–7 solve different bi-objective functions.

between the cost of energy supplied and the emission rate is not favourable, which indicates that the enhancement of one factor negatively impacts the outcomes of another. On the contrary, cases 2 and 3 show a better balance between themselves where the cost of energy supplied is optimized along with the VSI, and the cost of energy loss leads to even more distributed and effective solutions. In case 4, a highly smooth trade-off is observed between the objectives, which implies that it efficiently balances the cost of energy supplied and voltage deviation, and adjust-

ments to one objective function do not dramatically affect the performance of the other.

Figure 8 shows the final non-dominated solutions from case 5 to case 8. It illustrates a multifaceted view of the Pareto fronts comprising all five objective functions. Nevertheless, each case optimizes two conflicting objectives at a time. The visualization in Figure 8 emphasizes the proposed algorithm's ability to provide an all-inclusive overview of the trade-offs and synergies between different objective functions to achieve the desired goals.

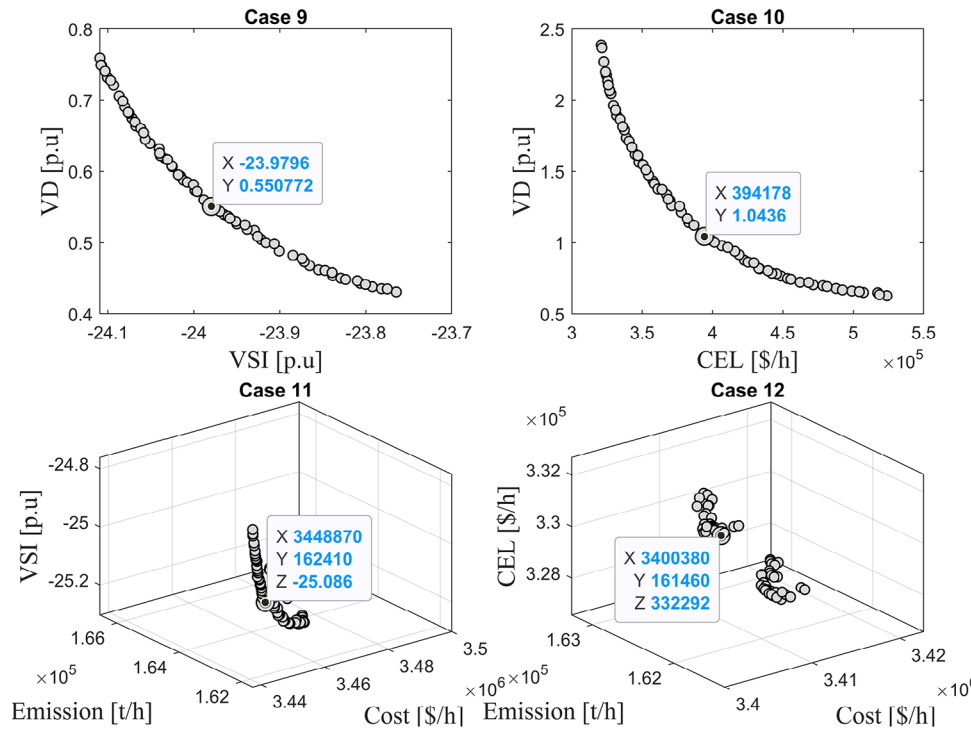


FIGURE 9 The final non-dominated solutions of cases 9–12 solve bi and tri-objective functions.

Figure 8 shows the ability of the proposed algorithm to balance conflicting objective functions in a complex distribution network efficiently. The 744 decision variables are considered across all cases, showing the complexity and scale of the posed problem. Figure 8 features the BCS [66], where the larger size marker is used to signify its importance, the best compromise solutions represent the balanced outcome among the obtained solutions, offering optimal values across all objective functions without bias. To identify and implement the best compromise solutions in the power system, a fuzzy set theory is employed to guide the selection process to obtain the most promising solution throughout a specific time span. Cases 7 and 8 show a smoother Pareto front, indicating a balanced trade-off between the objective functions. Case 5 and case 6 present non-smoother Pareto fronts, showing the influence of emission on the other objective function, which implies that the grid power supply impacts the emission rate. The minimization of emission rate harnesses more active and reactive power from renewable energy sources, including wind and solar-based DGs and BESS, and improves the overall performance of the distribution network.

Figure 9 gives an in-depth analysis of Cases 9 and 10, which handle two objective functions, while Cases 11 and 12 examine three different objective functions.

Figure 9 shows that in cases 9 and 10, the population members find a continuous Pareto front with better trade-offs among objective functions, that is, minimization of VD, VSI, and CEL. Population members are widely distributed in case 11, which comprises tri-objective functions, and the circle with darker markers shows the BCS among obtained solutions. The pop-

ulation members shown in case 12 find a discontinuous Pareto front, indicating gaps in the trade-off curves. The distribution of the population members indicates a range of trade-off solutions. The highlighted data point with a darker marker size represents the BCS, indicating the best trade-off solution according to a specific criterion.

Figure 10 shows the trade-offs among tri-objective Pareto fronts emphasizing the relationship between the emission rate and four other technical and economic objective functions, that is, cost of energy supplied, voltage deviation, voltage stability index, and cost of energy loss. In case 13, a continuous tri-objective PF is observed, whereas cases 14–16 give a wider distributed Pareto front representing diverse outcomes based on different objective functions. The analysis expanding from Figures 7–10 shows that the combination of all objective functions underlines the necessity to strike a balance among proposed technical, economic, and environmental objectives, especially in case 17, where all five objective functions are optimized simultaneously. This case provides an all-inclusive comparison that signifies how each objective function influences the siting and sizing of solar PV-type DGs, wind-based DGs, BESS, and power supplied by the main grid.

To emphasize the proposed algorithm's efficiency and superiority, Figure 11 illustrates a parallel plot comparing the final non-dominated solutions of the proposed framework with other contemporary MOEAs. The detailed comparison highlights the proposed framework's decision-making. It shows how it solved several objectives together while demonstrating its efficiency by generating well-rounded solutions in complex multi-objective optimization problems.

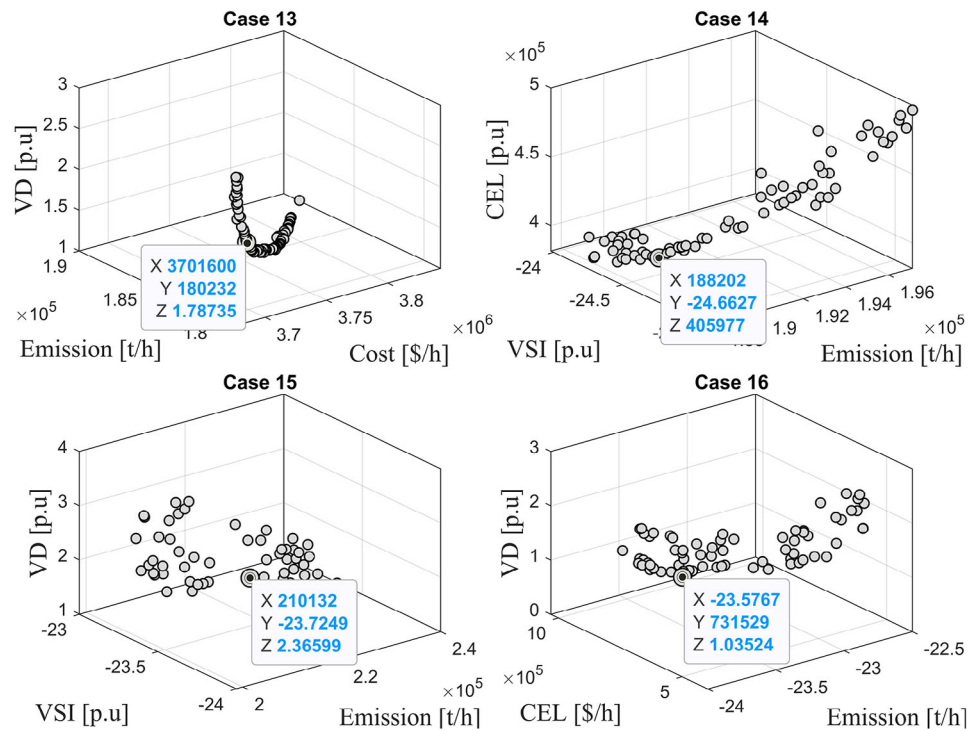


FIGURE 10 The final non-dominated solutions of cases 13–16 solve different tri-objective functions.

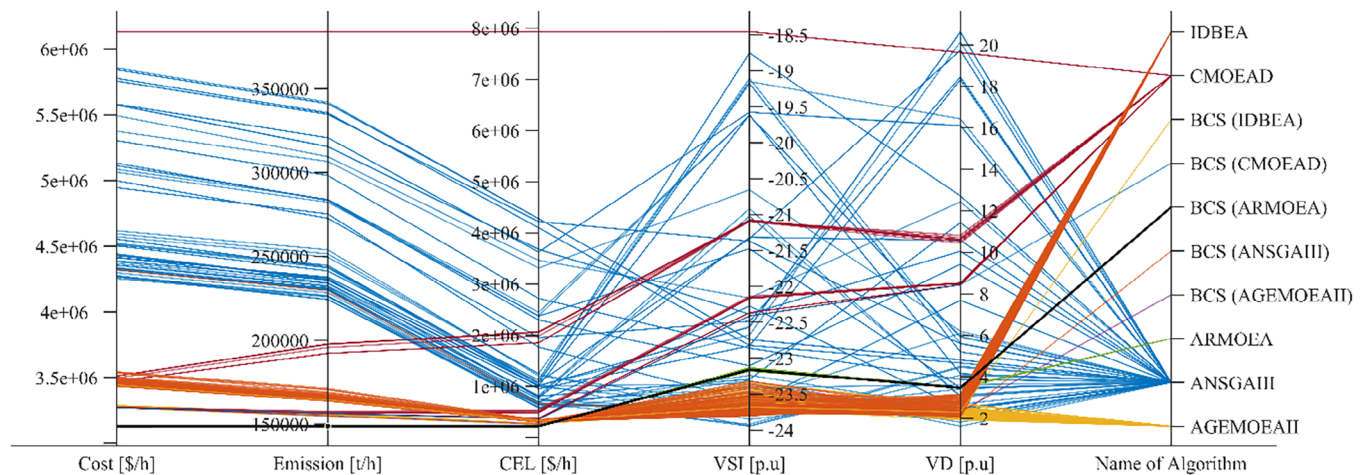


FIGURE 11 Comparison of various MOEAs for the solution of five conflicting objective functions.

The Figure 11 shows that the AGEMOEAI [16] and ARMOEA [67] are the best competitors to the proposed algorithm to optimize conflicting objective functions, whereas CMOEAD [68] and ANSGAIII [68] get stuck into the local optimal solutions. The widely distributed solutions obtained in ANSGAIII [68] shows that it is also stuck in the local minima, while in the proposed formulation, all the objective functions are minimized. The final Pareto front in a parallel plot with trade-offs among all objective functions is shown in Figure 12.

Figure 12 illustrates that the proposed framework finds the near global optimal solutions for all objective functions, and all

of the solutions are uniformly distributed. The decision maker can choose a single solution; however, this study employed fuzzy weight functions to find the best compromise solutions. Table 5 provides a statistical comparison between the proposed algorithm and different MOEAs regarding the BCS and the best and worst solutions for all objective functions.

Table 5 gives the statistical values of ANSGAIII, AGEMOEAI, ARMOEA, CMOEAD, and the proposed algorithm. These values denote the outcome of desired objective functions, where each technique is evaluated based on its best minimum (Best, min), worst maximum (Worst max), and BCS

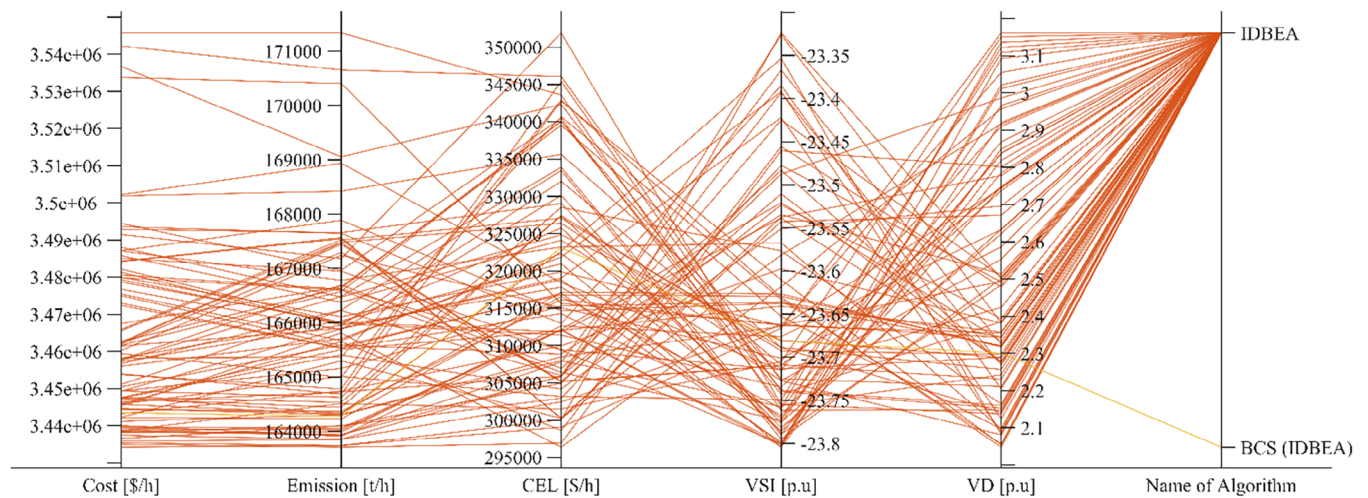


FIGURE 12 Final non-dominated solutions of all the objective functions.

TABLE 5 Statistical comparison of the proposed framework with the other recent MOEAs.

Parameters	Technique	Cost of energy supplied	Emission rate	CEL	VSI	VD
BCS	ANSGAIII	4.32×10^6	2.30×10^5	6.67×10^5	2.37×10^1	2.10×10^0
	AGEMOEAI	3.28×10^6	1.55×10^5	2.63×10^5	2.36×10^1	2.31×10^0
	ARMOEA	3.13×10^6	1.49×10^5	2.19×10^5	2.32×10^1	3.47×10^0
	CMOEAD	3.27×10^6	1.57×10^5	3.55×10^5	2.24×10^1	8.43×10^0
	Proposed	3.44×10^6	1.64×10^5	3.23×10^5	2.37×10^1	2.30×10^0
Best, min (OF)	ANSGAIII	4.25×10^6	2.24×10^5	5.73×10^5	2.39×10^1	1.61×10^0
	AGEMOEAI	3.27×10^6	1.55×10^5	2.53×10^5	2.37×10^1	1.92×10^0
	ARMOEA	3.13×10^6	1.49×10^5	2.18×10^5	2.32×10^1	3.45×10^0
	CMOEAD	3.27×10^6	1.56×10^5	3.55×10^5	2.24×10^1	8.43×10^0
	Proposed	3.43×10^6	1.64×10^5	2.96×10^5	2.38×10^1	2.05×10^0
Worst, max (OF)	ANSGAIII	5.86×10^6	3.42×10^5	4.28×10^6	1.87×10^1	2.06×10^1
	AGEMOEAI	3.29×10^6	1.56×10^5	2.77×10^5	2.33×10^1	2.69×10^0
	ARMOEA	3.13×10^6	1.49×10^5	2.21×10^5	2.31×10^1	3.50×10^0
	CMOEAD	6.13×10^6	3.84×10^5	7.93×10^6	1.85×10^1	1.96×10^1
	Proposed	3.55×10^6	1.71×10^5	3.52×10^5	2.33×10^1	3.16×10^0

values of each objective function. It can be noted from the values of Table 5 that in the category of BCS, the proposed framework demonstrates exceptional performance in terms of cost of energy supplies, emission rate, CEL, VSI, and VD by achieving values of 3.44×10^6 , 1.64×10^5 , 3.23×10^5 , 2.37×10^1 and 2.30×10^0 , respectively. The values indicate that the proposed algorithm achieves promising results across all objective functions as quantified by the BCS generated via fuzzy Wright functions. Table 5 shows the best and worst performance metrics of the other algorithms, which serves as a benchmark to check the efficiency of the proposed algorithm. In the worst-case scenario, the proposed algorithm shows substantial enhancements, especially compared to ANSGAIII and

CMOEAD algorithms, which give higher worst-case values across all objective functions. The values in Table 5 summarize that the proposed algorithm is efficient enough to tackle objective functions better than all other employed algorithms. Hence, the proposed algorithm is ideal for solving complex multi-objective optimization problems.

Table 6 gives the values of the decision variables obtained for all bi-, tri-, and penta-objective functions solved by the proposed algorithm.

Table 6 gives the optimal siting and sizing for PV-based DGs, wind-type DGs, and BESS across all 17 cases, highlighting the relationship among different objective functions successfully solved by the proposed algorithm. Each case shows how

TABLE 6 The proposed algorithm solves the optimal values of all objective functions.

Case	Description	No of OFs	PV _{site}	$\sum P_{SDG}$ [MWh]	WT _{site}	$\sum P_{WDG}$ [MWh]	BESST _{site}	$\sum P_{BESS_ch}$ [MWh]	$\sum P_{BESS_dch}$ [MWh]
1	C_E	2	14,17,30	15.685	8,14,25	43.327	31,32,31,20,16,2	36.810	-41.264
2	C_VSI	2	30,30,30	15.688	9,16,17	42.269	32,5,2,24,10,2	36.196	-41.099
3	C_CEL	2	6,6,30	15.534	8,15,30	41.434	7,25,5,8,5,26	38.561	-42.551
4	C_VD	2	7,14,25	15.279	8,14,30	42.787	18,7,22,8,7,2	39.195	-42.457
5	E_VSI	2	12,14,30	15.600	18,6,32	44.389	22,21,13,8,13,19	39.626	-42.449
6	E_CEL	2	30,30,30	15.756	14,30,30	44.468	2,15,24,30,5,3	39.058	-43.423
7	E_VD	2	25,30,30	15.353	3,25,30	44.288	18,8,14,10,2,2	37.444	-40.401
8	VSI_CEL	2	16,30,30	14.986	30,30,32	42.171	14,6,8,17,19,7	35.419	-37.297
9	VSI_VD	2	17,18,29	13.845	17,24,28	40.065	10,3,29,6,30,32	39.530	-36.877
10	CEL_VD	2	25,30,30	14.059	25,30,30	41.063	18,31,7,19,13,19	38.086	-37.018
11	C_E_VSI	3	6,6,18	15.631	6,6,7	44.139	27,32,3,32,26,15	34.872	-39.622
12	C_E_CEL	3	6,25,30	15.715	14,30,30	43.512	32,16,6,24,32,12	35.488	-40.498
13	C_E_VD	3	6,8,30	15.034	6,7,25	43.022	4,10,33,14,2,11	33.455	-37.965
14	E_VSI_CEL	3	14,8,30	14.019	5,15,30	40.983	11,7,30,7,7,2	34.669	-39.065
15	E_VSI_VD	3	9,18,30	14.350	14,14,29	40.420	18,19,7,26,7,2	34.643	-34.080
16	VSI_CEL_VD	3	25,30,30	14.155	7,7,30	40.746	25,15,29,30,7,7	33.142	-33.179
17	C_E_CEL_VSI_VD	5	6,12,30	15.163	7,25,30	42.540	7,2,7,31,6,2	37.314	-39.997

simultaneous optimization of two, three, and five objective functions influences the decision variables related to the location (site) and capacity (size) of the solar-based DGs, wind DGs, and battery energy storage systems. Cases 1–10 emphasize the bi-objective functions where each case simultaneously handles two objective functions. Cases 11–16 focus on optimizing three objective functions together to minimize them to achieve the desired results, and this step adds serious complexity to the algorithm's decision-making process to carry out optimal siting and sizing of the DGs and battery energy storage systems. Case 17 is a wide-ranging scenario in which all five objective functions are optimized simultaneously, showcasing the proposed algorithm's ability to handle such highly complex many-objective optimization problems. Table 6 notices the variations in optimal siting and sizing of distributed resources when there is a change in the combination of objective functions. Hence, the site and size parameters of PV-based DGs, wind-type DGs, and BESS are modified according to the specific objective functions given priority in the optimization process. Results obtained in Table 6 confirm that the proposed algorithm is a promising approach that solves all the technical, economic, and environmental objective functions to enhance the system's reliability and performance.

Figure 13 provides a thorough illustration by comparing the base case with other scenarios, emphasizing the influence posed by DG injections and BESS charging and discharging cycles.

From Figure 13, it can be noticed that during the spans of low demand, DGs play a significant role by injecting power to meet the load demand along with charging BESS devices simultaneously; this action not only fulfils the immediate load demand

of the system but also influences the overall performance of the load profile resulting in slight variations and obtaining a smooth load demand curve. Specifically, the power consumption from the grid is minimal during the time period of the low load demand, which typically occurs between 1 to 9 h. Nevertheless, it increases significantly during peak load demand. The BESS devices demonstrate their strategic operation by getting charged to maximum capacity during times of low demand and discharging themselves to support the grid during periods of high demand. The provided operational strategy efficiently shows the dynamic scheduling of DG and BESS devices by enhancing their performance in response to fluctuating load conditions. Figure 13 shows the contrast between power supplied by the grid in the base case and active power in all other cases, highlighting the efficiency due to the integration of DGs and BESS devices in lessening dependency on the main grid.

Besides the charging and discharging profiles of the BESS devices shown in Figure 13, focus on different techniques used to maintain balanced power levels. The employed strategies underscore how the BESS can mitigate load demand variations by making the energy system more efficient and reliable. Integrating DGs and BESS devices smooth the demand curve by reducing peak loads and ensuring a stable power supply throughout the day. It validates the potential of DG and BESS integration by enhancing grid resilience and underlining their essential role in enhancing the sustainability and efficiency of modern power networks. Moreover, Figure 14 compares the energy supplied by the grid, charging, and discharging of the BESS device across the whole horizon for all scenarios.

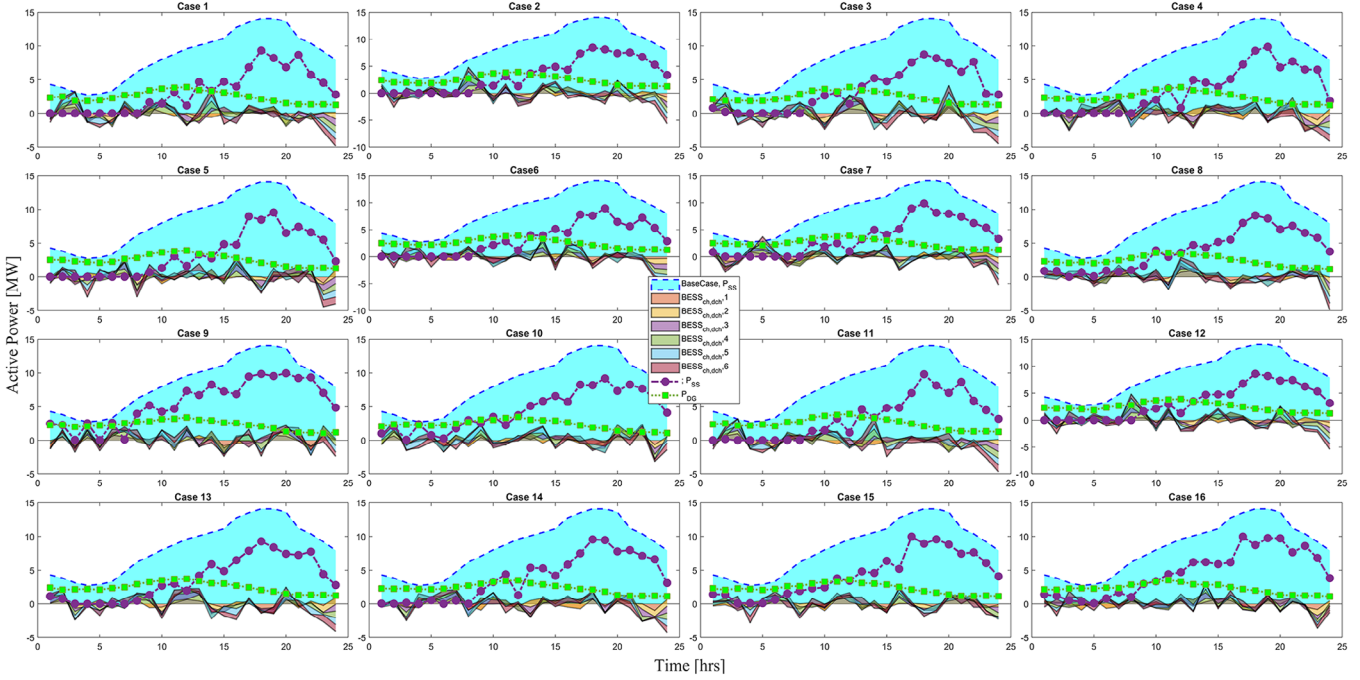


FIGURE 13 Comparison of the base case with all other cases.

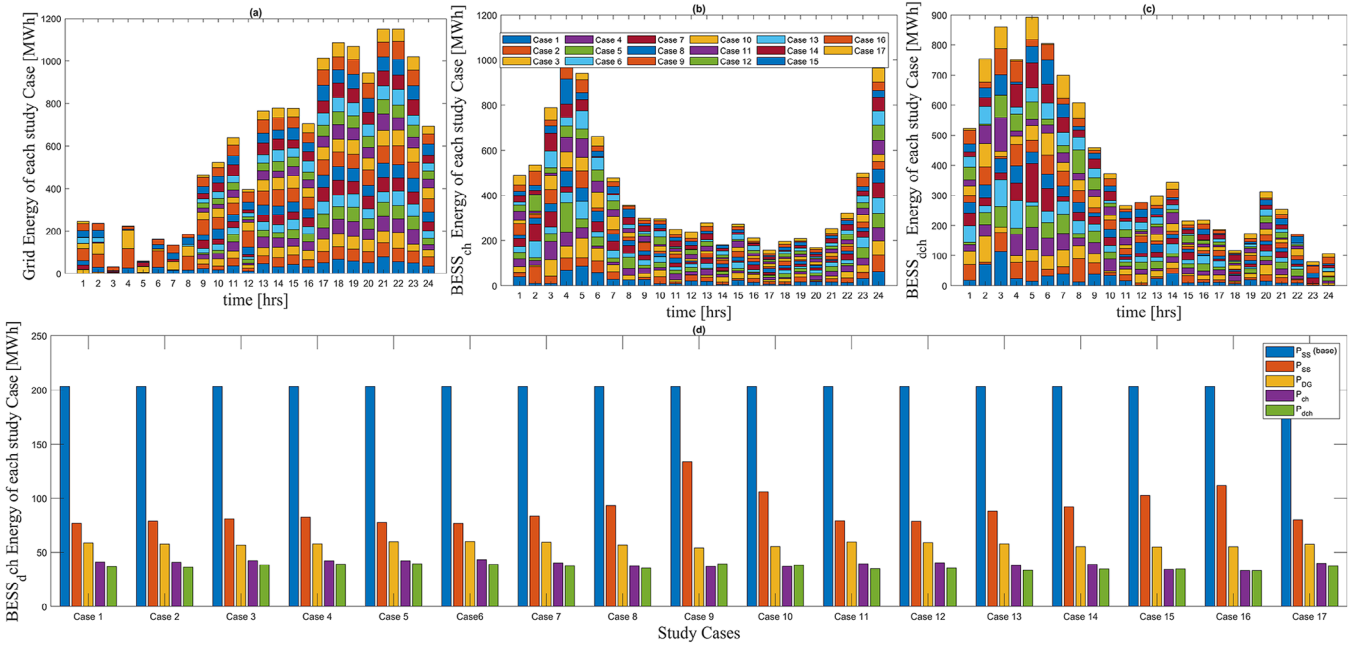


FIGURE 14 Comparison of (a) power supplied by the grid, (b) BESS charging power, (c) BESS discharging power, (d) MWh comparison of main grid, renewable energy based DGs, and charging and discharging energy of BESS for all 17 cases.

Figure 14a–c shows the pattern of energy supplied by the main grid and the charging and discharging of the BESS devices across all cases. During the low load demand, BESS devices are charged to maximum limits. In contrast, during the maximum load demand periods, BESS devices get discharged to flatten the load demand and enhance the reliability of the distribution

network. Figure 14d compares the active power supplied by the main grid in the base case to all other proposed cases. It can be noticed from Figure 14d that in the majority of the cases, the proposed algorithm reduced more than 60% of the power supply from the main grid except in case 9, where VSI and VD were the main objective functions. In case 9, the voltage set for all

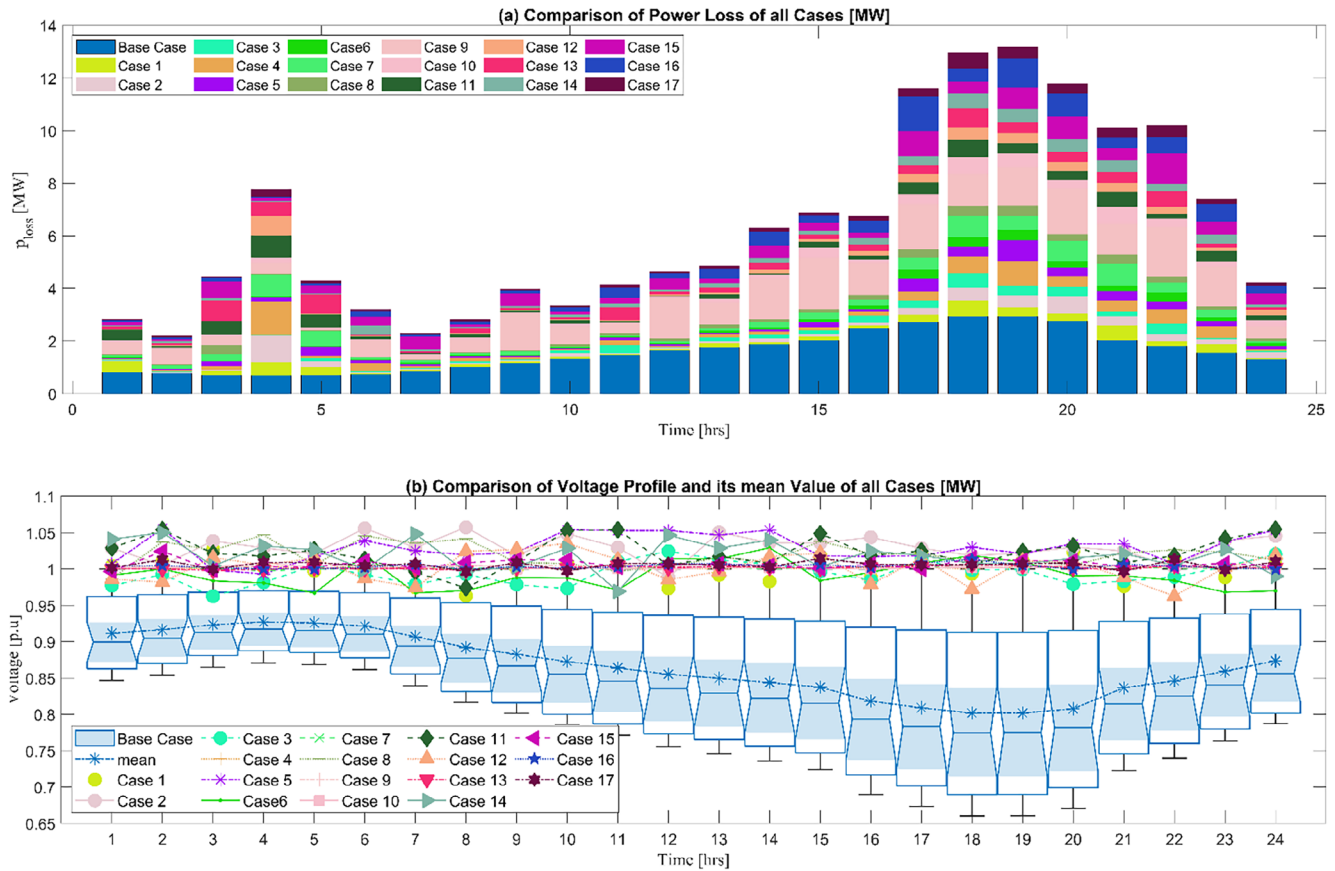


FIGURE 15 Comparison of (a) power losses and (b) voltage profiles of all cases.

buses is operated near the slack bus. Henceforth, more power is taken from the main grid along with DGs to enhance the voltage profile at the load buses. The optimal active power from DGs, charging, and discharging power of the BESS devices are obtained in case 17, where all technical, economic, and environmental objective functions were optimized simultaneously.

Figure 15 demonstrates the statistical comparison of power losses and voltage profile of the base case with all other cases.

Figure 15a illustrates the power losses across 24 h period. It can be noticed in Figure 15a, that the base case typically shows the highest power losses at various times, indicating its inefficiency in operating without suitable solar and wind DGs or integration of BESS devices. When comparing all cases, the active losses are higher in case 9 because it only minimized VD and VSI objective functions. It indicates that the power loss objective function should also be optimized along with VSI and VD because it is an important parameter that minimizes the total cost of the distribution network.

Figure 15b compares the voltage profiles in all cases with the base case. It plots voltage (p.u.) over 24 h with markers indicating their mean values. The base case shows more volatility and lower voltage levels, but introducing DGs and BESS devices into the distribution network improves the VSI across other cases, as evidenced by mean voltage levels. Case 9 shows the almost unity voltage (p.u.) across the time period due to the integration of DGs and BESS devices. The given graph-

ical representation highlights the benefits of DGs and BESS integration in reducing power losses and maintaining voltage stability. Table 7 gives simulation results of the base case, case 1, and case 2 across the entire time horizon.

The simulation results in Table 7 provide significant insight regarding the performance of different cases executed in this paper. In Table 7, case 6 shows substantial improvements by reducing the cost of energy supplied, emission rate, cost of energy loss, and cost of the power supplied by the main grid compared to the rest of the cases. The proposed algorithm shows the maximum VSI in case 5, which is the highest value compared to other cases throughout the study. Case 9 shows the lowest VD, indicating superior performance in maintaining the stable voltage across the network. While evaluating the performance of case 5, case 5, and case 9 compared to the other cases, these cases show exceptional performance by optimizing their respective objective functions. The proposed framework achieves tremendous improvements after an in-depth comparison of the results with the base case and other cases. Specifically, the maximum percentage savings were archived with a 49.6% reduction in costs of energy supplies, 62.2% reduction in emission rates, 92.3% reduction in the cost of energy loss, 91.9% improvement in voltage stability index, and an extraordinary 99.8953% reduction in voltage deviation in their respective best-performing cases.

TABLE 7 Simulation results of the base case, case 1, and case 2.

Case	Description	C	E	CEL	VSI	VD	C _{SS}	C _{DG}	C _{BESS}
0	Base case	6,607,541	416,549.4	3,090,858.1	8.970	526.4	6,607,541	—	—
1	C_E	3,344,449	157,849.1	394,711.6	21.867	10.132	2,878,100	755,195.4	288,846.7
2	C_VSI	3,394,762	161,994.6	404,003.9	26.073	20.521	2,939,906	742,550.6	287,695
3	C_CEL	3,430,560	165,891.5	304,022.0	21.457	8.718	2,998,206	730,207.3	297,853.5
4	C_VD	3,526,019	169,098.6	549,780.0	23.007	0.839	3,080,586	742,631.6	297,197.8
5	E_VSI'	3,405,735	159,316.6	373,155.1	26.182	20.713	2,936,210	766,669.5	297,145.2
6	E_CEL	3,327,168	157,634.4	238,905.8	21.779	9.112	2,861,176	769,949.8	303,958.4
7	E_VD	3,576,252	170,954.2	581,118.4	23.068	0.595	3,097,313	761,748.4	282,810.3
8	VSI_CEL	3,788,377	191,553.6	297,349.3	24.949	11.824	3,318,609	730,848.2	261,080.6
9	VSI_VD	4,938,188	274,183.9	190,6761.0	23.980	0.551	4,507,867	688,463.3	258,142.3
10	CEL_VD	4,056,421	216,702	394,178.5	23.228	1.044	3,611,903	703,647.4	259,129.2
11	C_E_VSI	3,448,869	162,409.5	538,272.1	25.086	17.380	2,962,087	764,133.9	277,351.7
12	C_E_CEL	3,400,380	161,459.9	332,291.6	22.234	9.228	2,925,996	757,866.8	283,483.1
13	C_E_VD	3,699,543	180,106.1	569,971.3	22.454	1.816	3,223,531	741,766.2	265,754.5
14	E_VSI_CEL	3,785,969	188,201.9	405,976.5	24.663	13.522	3,357,345	702,081.1	273,457.1
15	E_VSI_VD	4,094,402	210,132.3	810,268.7	23.725	2.366	3,632,670	700,289.5	238,557.4
16	VSI_CEL_VD	4,403,944	229,261.6	731,528.9	23.577	1.035	3,934,934	701,264.1	232,253.7
17	C_E_CEL_VSI_VD	3,443,498	164,277	323,089.1	23.681	2.301	2,985,542	737,934.5	279,979

5 | DISCUSSION

The presented study proposes the integration of wind and PV-based DGs with BESS devices in a complex distribution network. The study emphasizes the critical synergy between DGs and BESS devices to enhance the sustainability, dependability, and flexibility of the power system, which are the key elements to achieving a successful transition towards low-carbon energy systems. A novel multi-objective optimization framework is introduced in this paper to tackle many conflicting objective functions, that is, cost of energy supplied, emission rate, cost of energy loss, voltage stability index, and voltage deviation simultaneously. The proposed framework integrates normal boundary intersection and decomposition-based evolutionary algorithms, efficiently solving the siting and sizing of renewable-energy-based DGs and BESS devices. The proposed system considers the load demand variations, intermittency, and volatility of the renewable energy output. The standard IEEE 33 bus network was employed to validate the proposed algorithm. The results found in the study are compelling and showcase the significant improvement made by the proposed algorithm in enhancing the efficiency, environmental sustainability, and reliability of the distribution network. The results show a substantial reduction in power losses, voltage deviations, cost of energy supplied and energy loss, and emissions cost, along with significantly enhancing voltage stability.

When the results of the proposed algorithm are compared with other algorithms, it establishes its superiority over them throughout all cases, especially when multiple objectives are optimized simultaneously. Henceforth, the framework under-

lines its ability to perform the optimal siting and sizing of solar-type and wind-type DGs with BESS devices by reducing the need for grid station MVA capacity. The BESS devices show their importance by optimizing renewable energy utilization by stabilizing the grid and increasing the reliability of the power supply. The results enormously contribute to the ongoing research, making the distribution networks and smart grids more efficient and sustainable as they tackle economic, environmental, and technical objective functions by integrating DGs based on renewable energy and BESS devices. The accomplished research findings are thoroughly outlined in three distinct scenarios as given:

Scenario 1: It focuses on the DGs with a constant PQ load model, showing the capability of the proposed algorithm in optimizing their optimal capacities and locations. The presented cases validate the superiority of the proposed algorithm in mitigating the cost of energy supplies and energy loss, emission rates, and voltage deviations while enhancing the voltage stability index when compared to other contemporary algorithms considering both unity and lagging power factors.

Scenario 2: This scenario broadens the scope of the study by encompassing both DGs and SCs, underlining the proposed algorithm's ability to generate high-quality, non-dominated solutions. The achieved results validate the proposed algorithm's superiority in terms of convergence, diversity, and speed, emphasizing its robustness across a wide spectrum of objective functions.

Scenario 3: In this scenario, the proposed algorithm solves an extremely complex multi-period large-scale distributed allocation problem with many conflicting objective functions, further validating its superiority and efficiency in handling large-scale problems.

6 | CONCLUSION

This paper aims to integrate solar and wind-type distributed generators with battery energy storage systems simultaneously in a complex distribution network to enhance the reliability and flexibility of power systems. A novel many-objective optimization framework is introduced to solve multiple conflicting objective functions by reducing the cost of energy supplied, energy loss, voltage deviations, and emission rate while simultaneously enhancing the voltage stability index. The proposed approach successfully determines the optimal siting and sizing of the disturbed resources by exploiting improved normal boundary intersection and decomposition-based evolutionary techniques. The efficiency of the proposed algorithm is validated on the standard IEEE 33-bus network while considering the intermittency in the output of renewable energy sources and variations in load demand. The comparative analysis of the proposed algorithm with other contemporary algorithms indicates that it shows immense superiority over others in handling five conflicting objective functions, simultaneously ensuring its efficacy by mitigating the need for capacity requirements from the main station. The proposed framework achieved substantial savings of 49.6%, 62.2%, and 92.3% in energy supply costs, emission rate, and energy loss 92.3%, respectively, along with enormous enhancements of 91.9% in voltage stability index and 99.8953% in voltage deviations compared to the base case study.

7 | FUTURE WORK

Cutting-edge AI technologies can leverage the integration of wind, solar, and battery energy storages for the optimal siting and sizing of DERs, reducing energy losses and improving grid stability. In future work, implementing advanced machine learning algorithms could enhance the prediction accuracy of renewable energy sources and load demand. The reinforcement learning algorithms could optimize the battery energy storage systems operation in real-time considering the dynamic grid conditions. Moreover, the exploitation of deep learning frameworks can analyse complex datasets to find optimal locations for DGs based on geographical, environmental, and grid constraints.

NOMENCLATURE

Indices/Variables/Parameters

T_n	Span length of time
N_{DG}	Total number of DGs
nl	Total number of online feeders
N_b	Total number of buses
$b \in 1, \dots, N_b$	Index of a particular bus

N_{BESS}	Total Number of BESS
CB_L	Off line feeders
P_D, Q_D	Active and reactive load demand
C_{SS}, C_{DG}, C_{BESS}	Cost of MW supplied by substation, DG, and BESS
δ_{sr}	Voltage angle difference of branch between bus s and r
$\vec{x}$	Decision vector
$b(\vec{x})$	Equality constraint function
$g(\vec{x})$	Inequality constraint function
$c_i(\vec{x})$	i th degree of constraint violation
$CV(\vec{x})$	Overall constraint violation
$C(P_g)$	The total operating cost of thermal generators
p	Number of equality constraints
$q - p$	Number of inequality constraints
$site_{DG}, size_{DG}$	Optimal site and size of DG
$site_{BESS}, size_{BESS}$	Optimal site and size of BESS
$t \in 1, \dots, T_n$	Index of particular Time interval
$i \in 1, \dots, N_{DG}$	Index of generator
$j \in 1, \dots, N_{BESS}$	Index of particular BESS
f, k	Index of from and to bus
$l \in 1, \dots, nl$	Index of particular branch/feeder between f and t
$m \in 1, \dots, N_L$	Index of particular load bus
P_{SS}, P_{DG}	Active power supplied by SS and DG
P_{cb} and P_{dcb}	Charging and discharging power of BESS
N_L	Number of load buses
a	Direct cost parameter of SS
k_1, k_2, k_3	Quadratic cost parameters of DG and BESS
$C_{P_{loss}}, P_{loss}$	Cost of active power loss and power loss in the entire time span
V_{SS}, V_m	Voltage set point at the substation and m^{th} load bus
$SoC_{j(i\bar{m})}$	Initial state of charge
Q_{loss}	Reactive power loss
$Q_{SS}, Q_{SS}^{min}, Q_{SS}^{max}$	Actual, minimum, and maximum MVar of SS
$P_{SS}, P_{SS}^{min}, P_{SS}^{max}$	Actual, minimum, and maximum MW of SS
$P_{DG}, P_{DG}^{min}, P_{DG}^{max}$	Actual, minimum, and maximum MW of DG
$Q_{DG}, Q_{DG}^{min}, Q_{DG}^{max}$	Actual, minimum, and maximum MVar of DG
$V_{DG}, V_{DG}^{min}, V_{DG}^{max}$	Actual, minimum, and maximum voltage set point of DG
$V_m, V_m^{min}, V_m^{max}$	Actual, minimum, and maximum voltage of load bus
S_l, S_l^{max}	Actual and maximum MVA branch flow
A	Branch and bus incident matrix
SoC^t	State of charge at t^{th} time index
E_i^{min}, E_i^{max}	Minimum and maximum MWh capacity of BESS
P_{chi}^t and P_{dchi}^t	Charging and discharging power of BESS
r_l and x_l	Resistance and reactance of l th branch b/w bus f, k

- $G_{(mn)}$ Transfer conductance of branch between bus s and r
- S_{lq} Actual MVA branch flow limit

AUTHOR CONTRIBUTIONS

Ghulam Abbas: Conceptualization; writing—original draft; writing—review and editing; methodology. **Zhi Wu:** Conceptualization; supervision; writing—review and editing; funding acquisition; project administration; validation. **Aamir Ali:** Data curation; investigation; writing—review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data available on request from the authors.

ORCID

Ghulam Abbas  <https://orcid.org/0000-0001-7383-6798>

Zhi Wu  <https://orcid.org/0000-0003-4803-3920>

Aamir Ali  <https://orcid.org/0000-0002-1574-9060>

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